Optimisation Site Internet

Analyse et optimisation des performances web

Outils indispensables

- Outil Inspecter
- Elements
- Console
- Réseau
- Application

Confidential - Not for Public Consumption or Distribution

Elements

DevTools - www.lacommunautedesservices.fr/fr

Elements | Console | Sources | Network | Performance | Memory | Application | Security | Lighthouse

```
<!DOCTYPE html>
...<html lang="fr">  == $0
  ▸<head>…</head>
  ▾<body class="corps">
    ▾<o-app ng-version="5.2.11">
       <router-outlet></router-outlet>
      ▾<o-site>
        ▾<div class="site-content">
          ▸<o-nav _nghost-c1>…</o-nav>
          ▸<o-cgu>…</o-cgu>
          ▾<div class="page-content animated">
             <router-outlet></router-outlet>
            ▾<app-search _nghost-c6>
              ▸<div _ngcontent-c6 class="slider-wrapper">…</div>
              ▸<div _ngcontent-c6 class="container">…</div>
              </app-search>
            </div>
          ▸<o-gamification-notification _nghost-c2>…</o-gamification-notification>
          ▸<app-footer-nav _nghost-c3>…</app-footer-nav>
          </div>
        </o-site>
      </o-app>
      <script type="text/javascript" src="inline.812ba51….bundle.js"></script>
      <script type="text/javascript" src="polyfills.3882e16….bundle.js"></script>
      <script type="text/javascript" src="scripts.bf5e0f7….bundle.js"></script>
      <script type="text/javascript" src="vendor.0bc8351….bundle.js"></script>
      <script type="text/javascript" src="main.5bba57b….bundle.js"></script>
    ▸<iframe id="intercom-frame" style="position: absolute !important; opacity: 0
    !important; width: 1px !important; height: 1px !important; top: 0 !important;
    left: 0 !important; border: none !important; display: block !important; z-
    index: -1 !important; pointer-events: none;" aria-hidden="true" tabindex="-1"
    title="Intercom">…</iframe>
    </body>
  </html>
```

Styles | Computed | Event Listeners | DOM Breakpoints | Properties

Filter :hov .cls +

```
element.style {
}

html {                                          styles.4384…undle.css:9
  -webkit-tap-highlight-color: transparent;
  font-size: 62.5%;
  font-weight: 400;
}

html {                                          styles.4384…undle.css:5
  font-family: sans-serif;
  ~~-ms-text-size-adjust: 100%;~~
  ~~-webkit-text-size-adjust: 100%;~~
}

*, :after, :before {                            styles.4384…undle.css:5
  ~~-webkit-box-sizing: border-box;~~
  box-sizing: border-box;
}

html[Attributes Style] {
  -webkit-locale: "fr";
}

html {                                          user agent stylesheet
  display: block;
}

Pseudo ::before element

*, :after, :before {                            styles.4384…undle.css:5
  ~~-webkit-box-sizing: border-box;~~
  box-sizing: border-box;
}

Pseudo ::after element

*, :after, :before {                            styles.4384…undle.css:5
```

html

Console | What's New

Console

DevTools - www.lacommunautedesservices.fr/fr

Elements | Console | Sources | Network | Performance | Memory | Application | Security | Lighthouse

top | Filter | All levels

```
▸ t                                                                    main.5bba57b…bundle.js:1
key  18bdb77beed966e8f374                                              main.5bba57b…bundle.js:1
getNotifications                                                       main.5bba57b…bundle.js:1
▸ Object                                                               main.5bba57b…bundle.js:1
▸ ERROR ▸ V                                                            vendor.0bc8351…bundle.js:1
▸ WebSocket connection to 'wss://www.lacommunautedesservices.fr/sockjs-node/749/ucz4hc1p/websocket' failed: WebSocket  vendor.0bc8351…bundle.js:1
is closed before the connection is established.
[Intervention] Slow network is detected. See https://www.chromestatus.com/feature/5636954674692096 for more details. Fallback font will be  fr:1
used while loading: https://fonts.gstatic.com/s/raleway/v17/1Ptug8zYS_SKggPNyC0IT4ttDfA.woff2
▸ ERROR ▸ V                                                            vendor.0bc8351…bundle.js:1
Intercepted                                                            main.5bba57b…bundle.js:1
▸ Array(1)                                                             main.5bba57b…bundle.js:1
shop_id437                                                             main.5bba57b…bundle.js:1
region_id6                                                             main.5bba57b…bundle.js:1
▸ Invalid Host/Origin header                                           vendor.0bc8351…bundle.js:1
▸ [WDS] Disconnected!                                                  vendor.0bc8351…bundle.js:1
DATA================== ▸ Array(100)                                   main.5bba57b…bundle.js:1
all feeds ▸ Array(100)                                                 main.5bba57b…bundle.js:1
HOMEFEEDS ▸ Array(9)                                                   main.5bba57b…bundle.js:1
[Intervention] Slow network is detected. See https://www.chromestatus.com/feature/5636954674692096 for more details.  vendor.0bc8351…bundle.js:1
Fallback font will be used while loading: https://fonts.gstatic.com/s/raleway/v17/1Pt_g8zYS_SKggPNyCgSQamb1W0lwk4S4WjMDrMfJh1Zyc61YA.woff
10 MASON FINIF                                                         main.5bba57b…bundle.js:1
[Violation] Forced reflow while executing JavaScript took 32ms
MASON FINIF                                                            main.5bba57b…bundle.js:1
>
```

Console | What's New

Réseau (1/2)

DevTools - www.lacommunautedesservices.fr/fr

Elements | Console | Sources | Network | Performance | Memory | Application | Security | Lighthouse

Preserve log | Disable cache | Online

Filter | Hide data URLs | All | XHR | JS | CSS | Img | Media | Font | Doc | WS | Manifest | Other | Has blocked cookies | Blocked Requests

Name	Status	Type	Initiator	Size	Time	Waterfall
info?t=1598252888599	200	xhr	polyfills.3882e16....bund...	390 B	211 ms	
fr.json	200	xhr	polyfills.3882e16....bund...	2.8 kB	232 ms	
?_format=json	200	xhr	polyfills.3882e16....bund...	2.4 kB	392 ms	
categories?_format=json	200	xhr	polyfills.3882e16....bund...	1.6 kB	663 ms	
collect?v=1&_v=j83&a=978600921&t=pageview&_s=...	200	xhr	polyfills.3882e16....bund...	216 B	219 ms	
19028?_format=json	200	xhr	polyfills.3882e16....bund...	4.6 kB	718 ms	
list?_format=json	200	xhr	polyfills.3882e16....bund...	1.7 kB	383 ms	
?_format=json&page=0	200	xhr	polyfills.3882e16....bund...	8.4 kB	858 ms	
?_format=json&page=0	200	xhr	polyfills.3882e16....bund...	1.6 kB	869 ms	
?_format=json&page=0&gamification=true&notificatio...	200	xhr	polyfills.3882e16....bund...	1.7 kB	951 ms	
1?_format=json	200	xhr	polyfills.3882e16....bund...	1.1 kB	985 ms	
?_format=json	200	xhr	polyfills.3882e16....bund...	875 B	978 ms	
xhr_streaming?t=1598252892168	(canceled)	xhr	polyfills.3882e16....bund...	0 B	1.72 s	
homefeeds?_format=json	200	xhr	polyfills.3882e16....bund...	130 kB	2.13 s	
collect?t=dc&aip=1&_r=3&v=1&_v=j83&tid=UA-7829...	200	xhr	polyfills.3882e16....bund...	443 B	709 ms	
token	200	xhr	polyfills.3882e16....bund...	879 B	1.03 s	

20 / 73 requests | 179 kB / 3.7 MB transferred | 1.1 MB / 8.7 MB resources | Finish: 11.71 s | DOMContentLoaded: 3.67 s | Load: 3.67 s

Console | What's New

Confidential - Not for Public Consumption or Distribution

Réseau (2/2)

DevTools - www.lacommunautedesservices.fr/fr

Elements | Console | Sources | Network | Performance | Memory | Application | Security | Lighthouse

Preserve log | Disable cache | Online

Filter | Hide data URLs | All | XHR | JS | CSS | Img | Media | Font | Doc | WS | Manifest | Other | Has blocked cookies | Blocked Requests

1000 ms | 2000 ms | 3000 ms | 4000 ms | 5000 ms | 6000 ms | 7000 ms | 8000 ms | 9000 ms | 10000 ms | 11000 ms

Name

- info?t=1598252888599
- fr.json
- ?_format=json
- categories?_format=json
- collect?v=1&_v=j83&a=978600921&t=pageview&_s=1&dl=...777-1&_gi..
- 190287_format=json
- list?_format=json
- ?_format=json&page=0
- ?_format=json&page=0
- ?_format=json&page=0&gamification=true&notificatio..._tutorial,gamific.
- 1?_format=json
- ?_format=json
- xhr_streaming?t=1598252892168
- homefeeds?_format=json
- collect?t=dc&aip=1&_r=3&v=1&_v=j83&tid=UA-78296777...0590329.1..
- token

20 / 73 requests | 179 kB / 3.7 MB transferred | 1.1 MB / 8.7 MB resource

Headers | Preview | Response | Initiator | Timing | Cookies

General

Request URL: https://www.lacommunautedesservices.fr/api/bookmark/?_format=json&page=0
Request Method: GET
Status Code: 200 OK
Remote Address: 195.25.25.203:443
Referrer Policy: no-referrer-when-downgrade

Response Headers view source

Access-Control-Allow-Credentials: true
Access-Control-Allow-Headers: DNT,X-CustomHeader,Keep-Alive,User-Agent,X-Requested-With,
If-Modified-Since,Cache-Control,Content-Type,Authorization,X-CSRF-Token,x-pocd
Access-Control-Allow-Methods: GET, POST, OPTIONS, PUT, DELETE, PATCH
Access-Control-Allow-Origin: https://www.lacommunautedesservices.fr
Cache-Control: must-revalidate, no-cache, private
Connection: keep-alive
Content-Encoding: gzip

Console | What's New

Confidential - Not for Public Consumption or Distribution

Application

DevTools - www.lacommunautedesservices.fr/fr

Elements | Console | Sources | Network | Performance | Memory | Application | Security | Lighthouse

Application
- Manifest
- Service Workers
- Clear storage

Storage
- Local Storage
  - https://www.lacommuna
- Session Storage
- IndexedDB
- Web SQL
- Cookies

Cache
- Cache Storage
- Application Cache

Background Services
- Background Fetch
- Background Sync
- Notifications
- Payment Handler
- Periodic Background Sync
- Push Messaging

Filter

Key	Value
taxonomies	[{"tid":"169","uuid":"be2efbbe-4599-4a6d-81a6-3db93ade35e7","revision_id":"169","langcode":"en","vid":{
pusherTransportEncrypted	{"timestamp":1598252892998,"transport":"ws","latency":888}
results	[{"id":"41","uuid":"6182ca11-9dbd-41d4-ae2f-110d3586f41c","user_id":{"target_id":"98","url":"/api/user/98.
s2auth-19028	cm9qby5yYWJlbmFuYWhhcnlAcHJveGltaXR5YmJkby5mcjpPcmFuZ2UyMDIwIQ==
sessionCookie-19028	PLJL6rVjKn5g0168-cCIXYh11ie0o-MLsAXo7_upsKQ
lastUser	19028
FEK	VywtwtwnrC-22kkeegpwc1tqajoxF4kC-22==
intercom-state	{"app":{"features":{"anonymousInboundMessages":false,"googleAnalytics":false,"inboundMessages":false,
homeFeedsAll	[{"id":"1063","type":"forum","changed":"1598118999","comments_changed":"1598248254","content":{"id".
s2auser-19028	{"uid":"19028","uuid":"c6d06d46-ffb4-42e2-8213-65a2694ca8db","langcode":"en","preferred_langcode":".


```
▼ [{tid: "169", uuid: "be2efbbe-4599-4a6d-81a6-3db93ade35e7", revision_id: "169", langcode: "en",…},…]
  ▶ 0: {tid: "169", uuid: "be2efbbe-4599-4a6d-81a6-3db93ade35e7", revision_id: "169", langcode: "en",…}
  ▶ 1: {tid: "168", uuid: "94edea6c-f4a8-4cae-a758-ecac880f3567", revision_id: "168", langcode: "en",…}
  ▶ 2: {tid: "167", uuid: "3fc958ec-6a07-4449-8bd0-34d1aef64e87", revision_id: "167", langcode: "en",…}
  ▶ 3: {tid: "166", uuid: "1dc40a17-209c-4525-9e1c-a78715a23d26", revision_id: "166", langcode: "en",…}
  ▶ 4: {tid: "165", uuid: "f7a8c9c4-1512-47fc-9111-8dbf4ace6a1d", revision_id: "165", langcode: "en",…}
  ▶ 5: {tid: "164", uuid: "3988456a-1292-4712-aea1-a76a37a3a177", revision_id: "164", langcode: "en",…}
  ▶ 6: {tid: "163", uuid: "e434c185-6bfc-41d2-8a5e-704d7bf37346", revision_id: "163", langcode: "en",…}
  ▶ 7: {tid: "162", uuid: "d0366f49-aaaa-48ae-9ba8-a9eae6ea76cf", revision_id: "162", langcode: "en",…}
```


Console | What's New

Confidential - Not for Public Consumption or Distribution

Optimisations Taille

- Minimiser le poids des éléments (images, HTML, CSS, JS)
  - Redimensionner les images
  - Minifier
- Utiliser la mise en cache côté client

Confidential - Not for Public Consumption or Distribution

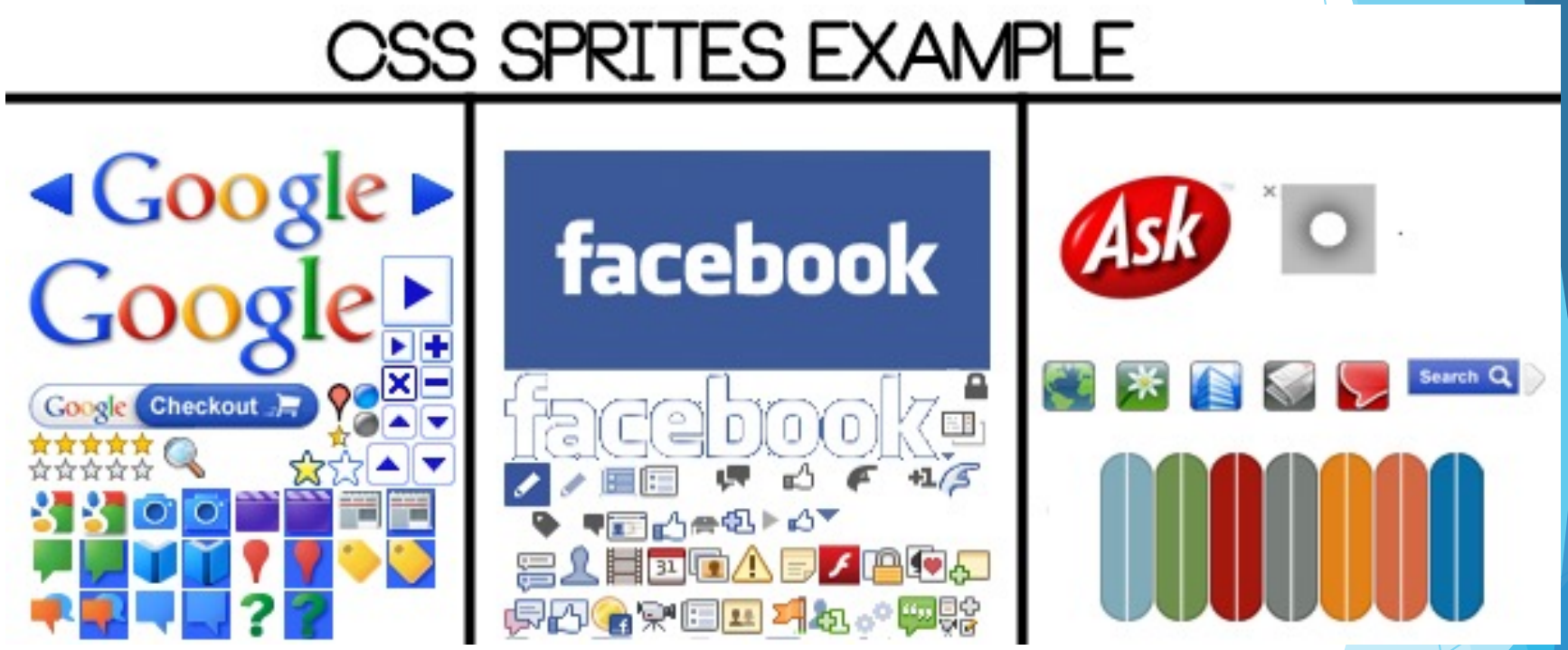
Optimisation CSS

- Concaténer les fichiers CSS

Confidential - Not for Public Consumption or Distribution

Optimisation CSS

- Utiliser des images sprites




```
#menugauche #edito{
  left:122px;
  background-position:-89px 0;
}
#menugauche #edito:hover,
#menugauche #edito:active,
#menugauche #edito:focus{
  background-position:-89px -52px;
}
```

t for Public Consumption or Distribution

Optimisation CSS

- Utilisation de CSS Critical Path pour charger les styles essentiels en premier

```
<head>
    <!-- CSS critique en ligne -->
    <style>
        body { font-family: Arial, sans-serif; }
        h1 { color: blue; }
    </style>

    <!-- Chargement différé du reste des styles -->
    <link rel="stylesheet" href="styles.css" media="print" onload="this
    <noscript><link rel="stylesheet" href="styles.css"></noscript>
</head>
```

- Outil pour identifier les CSS critiques
  - Google PageSpeed Insights (identifie les styles critiques)

Confidential - Not for Public Consumption or Distribution

Chargement asynchrone

- Charger en priorité les blocs visibles

```html
<head>
    <style>
        .header { background: #333; color: white; padding: 20px; }
        .hero { font-size: 2rem; text-align: center; }
    </style>
</head>
<body>
    <div class="header">Mon site</div>
    <h1 class="hero">Bienvenue sur mon site</h1>
    <!-- Le reste sera chargé plus tard -->
</body>
```

- Utiliser le lazy loading

```html
<img src="logo.png" alt="Logo" width="200"> <!-- Image prioritaire -->
<img src="image-lourde.jpg" loading="lazy" alt="Image en lazy loading"> <!-- Chargement différé -->
```

Confidential - Not for Public Consumption or Distribution

Optimisation des requêtes HTTP

- Utilisation du lazy loading pour les images et les vidéos.
- Utilisation de CDN pour charger des ressources statiques plus rapidement.
- Regroupement des requêtes API pour éviter des appels trop fréquents.

Confidential - Not for Public Consumption or Distribution

Optimisation Javascript

- Chargement différé de certains fichiers JavaScript avec defer ou async.
- Utilisation de tree shaking et code splitting pour ne charger que le code utile.
  - tree shaking : supprimer avec des outils (Webpack) les codes non utilisés
  - Code splitting : import dynamique à l'utilisation

```
import _ from 'lodash';

console.log(_.join(['Hello', 'World'], ' '));

document.getElementById('btnA').addEventListener('click', () => {
    import('./featureA').then(module => {
        module.featureA();
    });
});
```

Confidential - Not for Public Consumption or Distribution

Optimisation Serveur

- Compression Gzip du serveur Web
- Réutilisation des connexions HTTP avec Keep-Alive
- Optimisation du temps de réponse serveur avec un bon hébergement et une architecture adaptée (ex: serveur Nginx, Apache optimisé)
- Sécurisation avec HTTPS et HTTP/2 pour accélérer la communication entre client et serveur.
- Mise en place de cache avancé coté Serveur

Confidential - Not for Public Consumption or Distribution

SEO et Accessibilité

- Optimisation des balises title, meta, h1, h2
- Amélioration du temps de chargement mobile
- Vérification de l'accessibilité (contraste, lisibilité, navigation clavier)

Confidential - Not for Public Consumption or Distribution

Tests et Suivi des Performances

- Google PageSpeed Insight
- Lighthouse
- GTmetrix
- Analyse des logs serveur

Confidential - Not for Public Consumption or Distribution

Actions à réaliser

- Reprendre un site existant
- Appliquer les optimisations
- Documenter les changements
- Comparer les résultats

Confidential - Not for Public Consumption or Distribution